

Abd El Rahman Mohamed Karkar

AI Engineer | Machine Learning & Deep Learning

abdelrahmankarkar7@gmail.com | +20 106 864 8652 | Egypt | [linkedin.com/in/abd-el-rahaman-karkar](https://www.linkedin.com/in/abd-el-rahaman-karkar) | github.com/abdelrahmankarkar7-sys

SUMMARY

AI Engineer with hands-on experience designing and developing end-to-end Machine Learning and Deep Learning pipelines using Python, from data preparation and feature engineering to model training, validation, and deployment. Applied AI and ML techniques across Computer Vision, financial fraud detection, and multimodal AI-generated content detection, working with datasets ranging from 2,000+ samples to 50,000+ transaction records. Continuously strengthening industry-relevant AI/ML skills through applied projects and hands-on development.

EDUCATION

Delta University for Science and Technology, Egypt

Sep 2023 – Jul 2027

B.Sc. in Artificial Intelligence and Data Science

PROJECTS

Brain Tumor Detection Using Deep Learning

Python | Deep Learning | Computer Vision | Flask

- Trained and evaluated a CNN-based classification model on 3,000+ MRI scans, achieving approximately 92% classification accuracy.
- Preprocessed and augmented 3,000+ MRI scans using normalization, resizing, and data augmentation techniques to improve model generalization and reduce overfitting.
- Built and deployed an end-to-end Flask web application integrating image upload with real-time CNN-based MRI tumor classification.

Banking Fraud Detection & Loan Repayment Prediction

Python | Machine Learning | Scikit-learn | XGBoost

Python | Machine Learning | Scikit-learn | XGBoost

- Trained and validated fraud detection models using 50,000+ transaction records, achieving approximately 90% classification accuracy.
- Addressed class imbalance and assessed fraud detection model performance using precision, recall, and F1-score to ensure reliable results beyond accuracy alone.
- Built a loan repayment prediction component using financial data and applied feature engineering to prepare input variables for predictive modeling.

AI-Generated Content Detection System

Python | Machine Learning | Deep Learning | Flask | Scikit-learn | XGBoost

- Engineered a multimodal AI-generated content detection pipeline processing 2,000+ image, audio, and text samples, achieving approximately 85% classification accuracy.
- Combined modality-specific feature extraction across 3 content types—image, audio, and text—within a unified classification workflow.
- Packaged the multimodal detection pipeline behind a Flask API supporting real-time prediction across 3 input modalities: image, audio, and text.

EXPERIENCE

Freelance AI Engineer

Aug 2026 – Present

- Delivered 3 end-to-end Machine Learning and Deep Learning projects for clients, implementing workflows spanning data preprocessing, feature engineering, model development, evaluation, and deployment.
- Developed AI solutions across 3 domains—Computer Vision, financial fraud detection, and multimodal AI-generated content detection—using Machine Learning and Deep Learning techniques.
- Collaborated directly with clients to define project requirements, manage timelines, and deliver end-to-end AI solutions.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, Feature Engineering, Model Training & Evaluation

Deep Learning: TensorFlow, PyTorch, Keras, CNNs, RNNs, LSTMs, Transformers

AI Domains: Computer Vision, Natural Language Processing (NLP), OpenCV, Hugging Face

Data Analysis & Processing: Python, Pandas, NumPy, SQL, Exploratory Data Analysis (EDA), Data Preprocessing

Pipelines & Deployment: Data Preprocessing, Feature Engineering, Model Training & Evaluation, MLflow, Git, Docker, Flask

Data & Databases: SQL, Pandas, NumPy, MySQL, SQL Server, Exploratory Data Analysis (EDA)

Development Tools: Jupyter Notebook, google colab, vscode

LANGUAGES

Arabic — Native | English — Intermediate